

Shiyao Guo

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EDUCATION

Master of Engineering, Artificial Intelligence

December 2023

University of California, Los Angeles (UCLA) | Los Angeles, CA
GPA: 3.7/4.00

Bachelor of Science, Computer Science

2022

University of Rochester | Rochester, NY
Major GPA: 3.72/4.00; Cumulative GPA: 3.63/4.00; Dean's List in 2018, 2019, 2020, 2021, 2022

SKILLS

Programming Languages: Python, Java, JavaScript, Swift, C, C++, C#, MySQL, PostgreSQL, HTML, CSS
Frameworks & Libraries: Spring Boot, Flask, React, TensorFlow, PyTorch, Numpy, Pandas
Tools & Technologies: Docker, GitHub, Linux, Restful APIs

PROFESSIONAL EXPERIENCES

Walmart Inc.

Computer Vision Software Engineer | Dallas, TX

Feb 2025 – Current

- Designed and developed a scalable interactive web application for real-time camera monitoring and system diagnostics, leveraging **Python**, **Flask**, **React**, and **Cosmos DB** to ensure a seamless user experience
- Engineered and fine-tuned a deep learning pipeline using **YOLO** for real-time detection of humans and carts in dynamic retail environments; applied data augmentation and hyperparameter tuning to achieve 99.6% accuracy
- Developed a webpage using **React** and **Python** to project camera fields of view onto maps for visual coverage analysis

A Round Entertainment

Software Engineer Intern | Jersey City, NJ

Feb 2024 – Jan 2025

- Built a user-friendly dating website using **React** for frontend, while leveraging **Java** and **Spring Boot** for backend services
- Utilized **Hibernate** and **PostgreSQL** for efficient ORM mapping, enabling fast, scalable data operations

Qureator

Data Scientist Intern | San Diego, CA

Jun 2023 – Sep 2023

- Trained a Convolutional neural network in **TensorFlow** to extract features from bio-images and applied clustering to images
- Achieved a silhouette score of 0.7 in image clustering with well-defined clusters, providing valuable insights for biologists

Back to the Roots

Software Engineer Intern | Oakland, CA

Mar 2022 – Jun 2022

- Built a scalable **Flask** app using **Python**, **HTML**, **CSS**, and **React** to interact with Shippo API for real-time shipment tracking
- Implemented a robust Supply Request system using **Java**, **Spring Boot**, and **MySQL**

Costa Coffee

Machine Learning Engineer Intern | Remote

Jun 2021 – Aug 2021

- Utilized several models in **Python** including random forest, MLP, LSTM to predict the future sales based on many features
- Optimized LSTM using several techniques including gradient clipping, achieving a mean absolute percentage error of 4.2%

Milestone

Software Engineer Intern | Remote

Jun 2020 – Aug 2020

- Built Sales Force Automation software with **Python**, **JavaScript**, **HTML/CSS**, and **MySQL** to streamline performance tracking
- Created several webpages including login page and sales management page using **Python**, **JavaScript**, **CSS**, **HTML**, **MySQL**

PROJECTS

UCLA Capstone Project | San Diego Gas & Electric, Jun 2023 – Aug 2023

- Applied XGBoost, random forest, and multilayer perceptron to predict wildfire and attained accuracy of 0.75, 0.8, 0.75
- Led a group of students and determined which group of people are vulnerable to wildfire using feature importance analysis

Forum for music fans | University of Rochester, Feb 2022 – Apr 2022

- Built a forum using Python, HTML, CSS, JavaScript, MySQL, and AJAX where users can create subforum, post and comment